

MATLAB: Session 8

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Exercises

Part A – Diagonalizable Endomorphisms

A.1 Consider the diagonalizable endomorphism $f : \mathbb{R}^3 \rightarrow \mathbb{R}^3$, having a diagonalizing basis defined as

$$\mathcal{B} = \{v_1 = (1, 2, 0), v_2 = (0, 1, 1), v_3 = (0, 1, -1)\}$$

Moreover, suppose that 3 is an eigenvalue and $f(v_1) = f(v_2)$.

A.2 Is f injective? Is f surjective?

A.3 Assign to the variable **A1** the matrix representing f with respect to $\mathcal{B}$.

A.4 Assign to the variable **A2** the matrix representing f with respect to the canonical basis of $\mathbb{R}^3$.

A.5 Compute the dimension of $\ker f$ and assign it to the variable **dimkerf**.

A.6 Compute the dimension of $\operatorname{Im} f$ and assign it to the variable **dimimf**.

A.7 Compute a basis for $\ker f$ and assign it to the variable **basiskerf**.

A.8 Compute a basis for $\operatorname{Im} f$ and assign it to the variable **basisimf**.

Part B – Eigenvalues, Eigenvectors, Diagonalizability

B.1 Consider the endomorphism $h : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined as

$$h(x, y, z) = (x - y + z, 2y, z).$$

Assign to the variable **D** the matrix representing h with respect to the canonical basis of $\mathbb{R}^3$.

- B.2** Compute the characteristic polynomial of h and assign it to `charh`.
- B.3** Compute the eigenvalues of h and assign them to the column vector `eigh` (eigenvalues with algebraic multiplicity greater than one, must be repeated).
- B.4** Compute the eigenvectors of h . Save them as follows: if $v_1, \dots, v_j$ form a basis for the eigenspace relative to the eigenvalue $\lambda = a$, then assign them as columns of the matrix `Vh_a`.
- B.5** Is h diagonalizable? If yes, assign its diagonal form to the variable `Diagh`.
- B.6** If h is diagonalizable, compute the diagonalizing matrix and assign it to `R`.